SEMESTER S5

DATA COMPRESSION

(Common to CS/CD/CM/CR/AD/AI/AM/CN/CI)

Course Code	**PECST524**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Theory

Course Objectives:

1. To introduce students to basic applications, concepts, and techniques of Data Compression.
2. To develop skills for using recent data compression software to solve practical problems in a variety of disciplines.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Basic Compression Techniques :-** Data Compression Approaches - Variable-Length Codes, Run-Length Encoding, Space - Filling Curves, Dictionary-Based Methods, Transforms, Quantization. Huffman Encoding - Huffman Decoding, Adaptive Huffman Coding, Facsimile Compression. Run Length Encoding (RLE), RLE Text compression, Dictionary based Coding- LZ77, LZ78, LZW and Deflate: Zip and Gzip compression.	**10**
2	**Advanced Techniques :-** Arithmetic Coding - The Basic Idea, Implementation,Underflow; Image Compression- Introduction, Approaches to Image Compression, History of Gray Codes, Image Transforms, Orthogonal Transforms, The Discrete Cosine Transform, Intermezzo: Statistical Distributions, JPEG, Human Vision and Color, The Wavelet Transform, Filter Banks, WSQ, Fingerprint Compression	**10**

3	**Video Compression :-** Video Compression - Analog video, Digital Video, Motion Compensation. MPEG standards MPEG, H.261	**8**
4	**Audio Compression :-** Audio Compression - Companding, The Human Auditory System, Heinrich Georg Barkhausen, Linear Prediction, μ-Law and A-Law Companding, Shorten	**8**

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24 marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 subdivisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Describe the fundamental approaches in data compression techniques	**K2**
CO2	Illustrate various classical data compression techniques	**K3**
CO3	Illustrate various text and image compression standards	**K3**
CO4	Describe the video compression mechanisms to reduce the redundancy in video	**K3**
CO5	Understand the fundamental principles of audio data compression	**K2**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3										3
CO2	3	3	3									3
CO3	3	3	3									3
CO4	3	3	3									3
CO5	3	3										3

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	A Concise Introduction to Data Compression	David Salomon	Springer	1/e, 2008
2	Data compression: The Complete Reference	David Salomon	Springer	3/e, 2004
3	Introduction to Data Compression	Khalid Sayood	Morgan Kaufman	1/e, 2003

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Fractal and wavelet Image Compression techniques	Stephen Welstead,	PHI	1/e, 1999
2	Multimedia System	Sleinreitz	Springer	1/e, 2006
3	The Data Compression Book	Mark Nelson, Jean-loup Gailly	BPB Publications	1/e, 1996

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	An Introduction to Information Theory by Prof. Adrish Banerjee zt IIT Kanpur https://onlinecourses.nptel.ac.in/noc22_ee49/preview

SEMESTER 5

RESPONSIBLE ARTIFICIAL INTELLIGENCE

Course Code	**PEAIT527**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Theory

Course Objectives:

1. To master the methodologies for ensuring fairness, interpretability, and transparency in artificial intelligence systems
2. To evaluate and implement robust ethical frameworks, privacy-preserving techniques, and security controls in AI deployments.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Foundations of Responsible AI :-** Introduction to Responsible AI- Overview of AI and its societal impact; Fairness and Bias - Sources of Biases, Exploratory data analysis, limitations of a dataset, Preprocessing, in-processing, and post-processing to remove bias.	**7**
2	**Interpretability and explainability:-** Interpretability - Interpretability through simplification and visualization, Intrinsic interpretable methods, Post Hoc interpretability, Explainability through causality, Model-agnostic interpretation. Interpretability Tools - SHAP (SHapley Additive exPlanation), LIME(Local Interpretable Model-agnostic Explanations)	**10**
3	**Ethics, Privacy and Security:-** Ethics and Accountability -Auditing AI models, fairness assessment, Principles for ethical practices. Privacy preservation - Attack models, Privacy-preserving Learning, Differential privacy-Working, The Laplace Mechanism, Introduction to Federated learning. Security - Security in AI Systems, Strategies for securing AI systems and protecting against adversarial attacks	**10**

4	**Future of Responsible AI and Case Studies: -** Future of Responsible AI - Emerging trends and technologies in AI ethics and responsibility. Case Studies - Recommendation systems, Medical diagnosis, Computer Vision, Natural Language Processing.	9

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24 marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 subdivisions. **(4x9 = 36 marks)**	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Identify and describe key aspects of responsible AI such as fairness, accountability, bias, and privacy.	**K2**
CO2	Describe AI models for fairness and ethical integrity.	**K2**
CO3	Understand interpretability techniques such as simplification, visualization, intrinsically interpretable methods, and post hoc interpretability.	**K2**
CO4	Comprehend the ethical principles, privacy concerns, and security challenges involved in AI development and deployment.	**K3**
CO5	Understand responsible AI solutions for practical applications, balancing ethical considerations with model performance.	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	-	-	-	-	-	-	-	-	3
CO2	3	3	3	-	-	-	-	-	-	-	-	3
CO3	3	3	3	-	-	-	-	-	-	-	-	3
CO4	3	3	3	-	-	-	-	-	-	-	-	3
CO5	3	3	3	-	-	-	-	-	-	-	-	3

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Responsible Artificial Intelligence: How to Develop and Use AI in a Responsible Way	Virginia Dignum	Springer Nature	1/e, 2019
2	Interpretable Machine	Christoph Molnar	Lulu	1/e, 2020

	Learning			

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	ResponsibleAI: Implementing Ethical and Unbiased Algorithms	Sray Agarwal, Shashin Mishra	Springer Nature	1/e, 2021

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://youtu.be/3-xhMXeYIcg?si=x8PXrnk0TabaWxQV
2	https://youtu.be/sURHNhBMnFo?si=Uj0iellJs3oLOmDL [SHAP and LIME] https://c3.ai/glossary/data-science/lime-local-interpretable-model-agnostic-explanations/ https://shap.readthedocs.io/en/latest/ https://www.kaggle.com/code/bextuychiev/model-explainability-with-shap-only-guide-u-need
3	https://www.youtube.com/live/DA7ldX6OIG4?si=Dk4nW1R1zi_UMG_4
4	https://youtu.be/XlYhKwRLerc?si=IeU7C0BLhwn9Pvmi Case Studies https://www.kaggle.com/code/teesoong/explainable-ai-on-a-nlp-lstm-model-with-lime https://www.kaggle.com/code/victorcampelo/using-lime-to-explaining-the-preditions-from-ml

SEMESTER 5

MALWARE DETECTION AND ANALYSIS

Course Code	**PEAIT528**	**CIE Marks**	**40**
Teaching Hours/Week (L: T:P: R)	**3:0:0:0**	**ESE Marks**	**60**
Credits	**3**	**Exam Hours**	**2 Hrs 30 Mins**
Prerequisites (if any)	**PEAIT528**	**Course Type**	**Theory**

Course Objectives:

1. To develop a good understanding of malware analysis.
2. To identify different malware detection techniques.
3. To gain knowledge about advanced static malware analysis environment.
4. To gain knowledge about advanced dynamic malware analysis environment.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	INTRODUCTION: Introduction to Malware, Operating System Security Concepts, Malware Threats, Evolution of Malware, Malware Types - Viruses, Worms, Rootkits, Trojans, Bots, Spyware, Adwares, Logic Bombs, Malware Analysis, Static Malware Analysis, Dynamic Malware Analysis, Categories of Malware, Patterns of Malware Attack, Classification of Malware By Malicious Behaviors, Malware Attack Life Cycle.	9
2	MALWARE DETECTION TECHNIQUES: Signature-based techniques: malware signatures, packed malware signature, metamorphic and polymorphic malware signature Non-signature-based techniques: similarity-based techniques, machine-learning methods, invariant inferences.	9
3	ADVANCED STATIC ANALYSIS: X86 Disassembly: Levels Of Abstraction, Reverse-Engineering, X86 Architecture: Main Memory, Instructions, Opcodes And Endianness, Operands, Registers, Simple Instructions, The Stack, Conditionals, Branching, Rep Instructions, C Main Method And Offsets, Analyzing Malicious Windows Programs, Portable Executable File Format, Disassembling Malicious Executable Programs. Malware Components, Malware Distribution Mechanisms, YARA.	12

4	ADVANCED DYNAMIC ANALYSIS: Debugging Malware, Ollydbg, Kernel Debugging With Windbg, Setting Virtual Environments-Sandboxes, Emulators, Hypervisors, Virtual Machines, Live Malware Analysis, Dead Malware Analysis, Analyzing Traces Of Malware-Systemcalls, Api Calls, Registries, Network Activities. Debugging Tricks For Unpacking Malware.	10

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	To develop a good understanding of malware analysis.	K2
CO2	To identify different malware detection techniques.	K2
CO3	To gain knowledge about advanced static malware analysis environment.	K3
CO4	To gain knowledge about advanced dynamic malware analysis environment.	K3

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	√	√	√	√	-	-	-	-	-	-	-	√
CO2	√	√	√	√	√	-	-	-	-	-	-	√
CO3	√	√	√	√	-	-	-	-	-	-	-	√
CO4	√	√	√	√	√	-	-	-	-	-	-	√

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Practical Malware Analysis: The hands-On Guide to Dissecting Malicious Software	Michael Sokorski, Andrew Honig	No Starch Press,	2/e, 2012
2	Malware Analysis and Detection Engineering: Comprehensive Approach to detect and Analyze Modern Malware.	Abhijith Mohanta, Anoop Saldanha	Apress	1/e, 2020
3	Malware Detection	Christodorescu, M, Jha, S. Maughan, D. Song, D. Wang,C	Springer US	1/e, 2007

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Practical Reverse Engineering	Dang, Gazet, Bachaalany	Wiley	2014
2	MalwareForensics: Investigating and Analyzing Malicious Code	Cameron H. Malin, Eoghan Casey, and James M. Aquilina	Syngress Publications	1st Edition, 2008

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
3	https://www.cybrary.it/course/malware-analysis/
3	https://www.udemy.com/course/malware-analysis-and-reverse-engineering/

SEMESTER 5

UI & UX Design

Course Code	PEAIT525	CIE Marks	40
Teaching Hours/Week (L:T:P: R)	3:0:0:0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs 30 Min
Prerequisites (if any)	NA	Course Type	Theory

Course Objectives:

1. To introduce the fundamental concepts, principles and processes of UI and UX design.
2. To develop an understanding of user-centered design, design thinking and UX research methodologies.
3. To familiarize students with UI design principles, interaction patterns, branding and responsive interface design.
4. To provide hands-on experience in wireframing, prototyping, usability testing and iterative design using industry-standard tools.
5. To enable students to design intuitive and user-friendly digital interfaces by applying UI/UX design principles.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**FOUNDATIONS OF DESIGN** UI vs. UX Design - Principles of Good Design, Human-Centered Design, Core Stages of Design Thinking, Divergent and Convergent Thinking, Brainstorming and Gamestorming, Observational Empathy, Personas and User Journey Maps, Information Architecture.	9
2	**FOUNDATIONS OF UI DESIGN** Visual and UI Principles - Visual Design Principles, Typography, Color Theory and Accessibility, Icons and Imagery, UI Elements and Components, UI Patterns - Interaction Behaviors and Principles – Navigation Design, Responsive and Adaptive Design, Branding - Style Guides and Design System.	8
3	**FOUNDATIONS OF UX DESIGN** Introduction to User Experience - Why You Should Care about User Experience - Understanding User Experience - Defining the UX Design Process and its Methodology - Research in User Experience Design - Tools and Method used for Research - User Needs and its Goals - Know about Business Goals.	10

4	**WIREFRAMING, PROTOTYPING AND TESTING** Sketching Principles - Sketching Red Routes - Responsive Design – Wireframing - Creating Wireflows - Building a Prototype - Building High-Fidelity Mockups - Designing Efficiently with Tools - Interaction Patterns - Conducting Usability Tests - Other Evaluative User Research Methods - Synthesizing Test Findings - Prototype Iteration.	**10**

Suggested Lab Experiments for Practice

1. Designing a Responsive layout for a societal application.
2. Exploring various UI Interaction Patterns.
3. Developing an interface with proper UI Style Guides.
4. Developing Wireflow diagram for application using open source software.
5. Exploring various open source collaborative interface platforms.
6. Hands on Design Thinking Process for a new product.
7. Brainstorming feature for proposed product.
8. Defining the Look and Feel of the new Project.
9. Create a Sample Pattern Library for that product (Mood board, Fonts, Colors based on UI principles).
10. Identify a customer problem to solve.
11. Conduct end-to-end user research - User research, creating personas, Ideation process (User stories, Scenarios), Flow diagrams, Flow Mapping.
12. Sketch, design with popular tools and build a prototype and perform usability testing and identify improvements.

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	***Internal Ex***	***Evaluate***	***Analyse***	***Total***
5	**15**	**10**	**10**	**40**

Criteria for Evaluation(Evaluate and Analyse): 20 marks

Assignment evaluation pattern:

1. Correctness and Accuracy (30%) - Correct Solution and Implementation.
2. Effectiveness and Efficiency (25%) - Algorithm Efficiency and Performance Metrics.
3. Analytical Depth (25%) - Problem Understanding and Solution Analysis.
4. Justification and Comparisons (20%) - Choice Justification and Comparative Analysis.

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks. **(8x3 =24 marks)**	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 3 subdivisions. Each question carries 9 marks. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Build UI for user Applications	K3
CO2	Evaluate UX design of any product or application	K4
CO3	Demonstrate UX Skills in product development	K3
CO4	Implement Sketching principles	K3
CO5	Describe Wireframe and Prototype	K2

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	3	-	-	-	-	-	-	-	-
CO2	2	3	2	3	2	-	-	-	2	2	-	2
CO3	2	3	3	2	2	-	-	-	-	-	-	2
CO4	2	2	3	-	-	-	-	-	-	-	-	-
CO5	2	2	3	-	-	-	-	-	-	-	-	-

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	UX for Beginners	Joel Marsh,	O'Reilly ,	1/e, 2022
2	Laws of UX using Psychology to Design Better Product & Services	Jon Yablonski,	O'Reilly	2/e, 2021

Reference Books

Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Designing Interface	Jenifer Tidwell, Charles Brewer, Aynne Valencia,	O'Reilly	3/e, 2020
2	Refactoring UI	Steve Schoger, Adam Wathan	Ebook	1/e, 2018
3	Don't Make Me Think, Revisited: A Commonsense Approach to Web & Mobile	Steve Krug	Ebook	3/e, 2015

Video Links (NPTEL, SWAYAM)	
Module No.	**Link ID**
1	https://nptel.ac.in/courses/124107008
2	https://www.youtube.com/@acciojob
3	https://www.youtube.com/@freecodecamp

SEMESTER 5

AUGMENTED AND VIRTUAL REALITY

Course Code	**PEAIT595**	**CIE Marks**	**40**
Teaching Hours/Week (L: T:P: R)	**3:0:0:0**	**ESE Marks**	**60**
Credits	**5/3**	**Exam Hours**	**2 Hrs. 30 Min.**
Prerequisites (if any)	**None**	**Course Type**	**Theory**

Course Objectives:

1. To introduce the fundamental concepts, components, and technologies of Augmented Reality (AR) and Virtual Reality (VR).
2. To familiarize students with the development of AR and VR applications using appropriate tools, interaction techniques, and user experience (UX) principles.
3. To provide an understanding of the hardware, software architectures, rendering techniques, and performance considerations in AR and VR systems.
4. To explore the applications, challenges, ethical considerations, and emerging trends in immersive technologies.

SYLLABUS

Module No.	**Syllabus Description**	**Contact Hours**
1	**Introduction to Augmented and Virtual Reality** - Overview of AR, VR and Extended Reality (XR): Definitions, History and Evolution; Basic Concepts: Immersion, Presence, Interaction, Spatial Computing; Components of AR and VR Systems: Display, Input, Tracking, Sensors; AR vs. VR vs. Mixed Reality (MR) Applications: Gaming, Education, Healthcare, Manufacturing, Retail, Entertainment.	**9**
2	**AR and VR Application Development -** AR SDKs and Frameworks: ARCore, ARKit, WebXR (Introduction), VR Development using Unity and Unreal Engine; Interaction Techniques: Gesture Recognition, Voice Interaction, Eye Tracking, Haptic Feedback; UX/UI Design Principles for Immersive Environments, Scene Design and Asset Integration; Case Studies of Successful AR/VR Applications.	**9**
	AR and VR Hardware and Software - AR Hardware: Smart Glasses, Head-Up Displays, Mobile Devices; VR Hardware: Head-Mounted Displays, Motion Controllers, Tracking Systems. Software	

3	Architecture for AR/VR Systems; Rendering Techniques: Real-Time Rendering, 3D Graphics, Spatial Audio, Physics Engines. Performance Optimization: Latency Reduction, Frame Rate Optimization, Occlusion Culling.	**9**
4	**Advanced Topics and Applications in AR and VR -** Challenges in AR and VR: Technical, Ethical and Social Issues AR/VR in Education, Training, and Simulation AR/VR in Healthcare, Industrial Training, and Digital Twins; Future Trends: Mixed Reality (MR), Augmented Virtuality (AV), Extended Reality (XR), Metaverse; Responsible XR: Privacy, Security, Accessibility, and Societal Impact.	**9**

Sample Practical questions :

1. Develop an AR application to detect a horizontal plane and place a 3D object at the detected location.
2. Create a VR environment that enables users to navigate using teleportation.
3. Design an interactive VR scene where users can pick up and manipulate virtual objects.
4. Create an image-tracking AR application to display a 3D model corresponding to a given image marker.
5. Develop a simple virtual museum with at least three exhibits and interactive information panels.
6. Design an immersive educational scene incorporating lighting, textures, and spatial audio.
7. Build an AR application for visualizing a real-world object (e.g., furniture, machine component, or anatomical model).
8. Evaluate an existing AR/VR application and prepare a report discussing its usability, interaction techniques, performance, and user experience.

Course Assessment Method (CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	***Internal Ex***	***Evaluate***	***Analyse***	***Total***
5	**15**	**10**	**10**	**40**

Criteria for Evaluation (Evaluate and Analyze): 20 marks

Assignment evaluation pattern:

1. Correctness and Accuracy (30%) - Correct Solution and Implementation.
2. Effectiveness and Efficiency (25%) - Algorithm Efficiency and Performance Metrics.
3. Analytical Depth (25%) - Problem Understanding and Solution Analysis.
4. Justification and Comparisons (20%) - Choice Justification and Comparative Analysis.

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24 marks)**	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 3 subdivisions. Each question carries 9 marks. **(4x9 = 36 marks)**	**60**

question out of two questions.

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the fundamental concepts, components, technologies, and applications of Augmented Reality (AR) and Virtual Reality (VR).	K2
CO2	Apply AR/VR development tools, interaction techniques and user-centered design principles to develop basic immersive applications.	K3
CO3	Implement AR/VR solutions by utilizing appropriate hardware, software architectures, rendering techniques and performance optimization methods.	K3
CO4	Analyze AR/VR applications with respect to usability, performance, ethical considerations and emerging trends to recommend suitable solutions for real-world scenarios.	K4

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	-	-	2	-	-	-	-	-	-	-
CO2	2	-	3	3	2	-	2	-	-	-	-	-
CO3	2	2	3	3	2	-	-	-	-	-	-	-
CO4	2	-	-	-	3	2	2	2	-	-	-	-

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Augmented Reality: Principles and Practice	Dieter Schmalstieg, Tobias Hollerer	Addison-Wesley Professional	1/e, 2016
2	Learning Virtual Reality: Developing Immersive Experiences and Applications for Desktop, Web, and Mobile	Tony Parisi	O'Reilly Media	1/e, 2015

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Augmented Reality and Virtual Reality: The Power of AR and VR for Business	Timothy Jung, M. Claudia tom Dieck	Springer	1/e, 2021
2	Virtual Reality and Augmented Reality: Myths and Realities	Bruno Arnaldi, Pascal Guitton, Guillaume Moreau	CRC press	1/e, 2018
3	Practical Augmented Reality: A Guide to the Technologies, Applications, and Human Factors for AR and VR	Steve Aukstakalnis	Addison-Wesley Professional	1/e, 2017

Video Links (NPTEL, SWAYAM…)	
No.	**Link ID**
1	https://onlinecourses.swayam2.ac.in/nou23_ge34/preview
2	https://www.youtube.com/watch?v=zLMgdYI82IE
3	https://www.youtube.com/watch?v=MGuSTAqlZ9Q